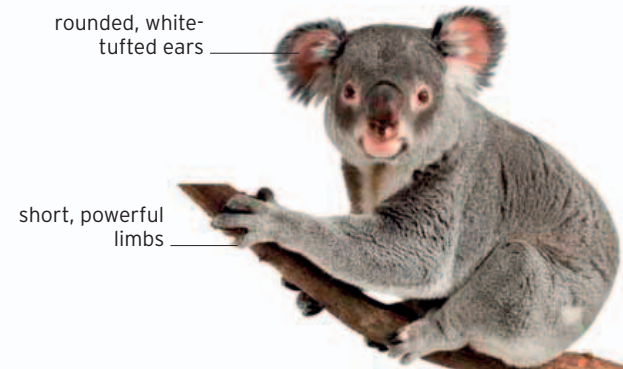


Koala

Phascolarctos cinereus



Koalas are marsupials that have evolved the unique ability to eat a plant that would poison other animals. They feed almost exclusively on eucalyptus leaves, but not just any eucalyptus will do. Koalas are choosy eaters, feeding only on a few of the 600-plus eucalyptus species found in Australia. They also avoid eating leaves from trees growing in poor soil because these are the most toxic.

Special adaptations

A fiber-digesting organ known as the cecum helps koalas to feed on eucalyptus without ill effects. In humans, this small pouch at the beginning of the large intestine is about 3 in (6.25 cm) long, but in koalas, it reaches 80 in (200 cm). The cecum contains millions of bacteria that break down the leaf fiber and toxic oils and enable absorption of nitrogen. A slow metabolic rate keeps food within the digestive system for extended periods. Even so, a koala absorbs only about 25 percent of what it consumes—so one animal must eat up to 1 lb (up to 500 g) of eucalyptus leaves a day in order to survive.

Fortunately, the koala's teeth are made for the job. Sharp front incisors cut leaves from stems, while molars shear and crush, breaking the food down before it reaches the cecum. While it chews, a koala stays safely anchored in a tree to avoid ground predators such as dogs. As well as sharp, curved claws, its paws have rough pads that grasp bark and branches. The front

paws have two digits that oppose the other three to give the animal a firm, fistlike grip. The koala's dense, thick coat varies from light gray in northern animals to dark-brown in southern ones. The soft, long, fur protects the koala from extreme temperatures as well as rain.

Mainly nocturnal, koalas are highly territorial animals; each adult maintains a home range within a breeding group, marking the trees it visits regularly with scratches. The males also scent-mark from a brown gland in the middle of their chests, which they rub on the base of trees.

Joey in a pouch

The breeding season lasts from August to February, punctuated by frequent bellowing from males. Females give birth to a single hairless, embryonic joey, roughly 1 in (2 cm) long and weighing less than a gram. The tiny joey climbs from the birth canal to its mother's pouch and once inside, latches on to one of her two teats. It stays there and develops, for about 22–30 weeks, when it begins to feed on “pap” as well as milk. Pap, a special type of dropping the mother produces, contains the microbes needed to digest eucalyptus leaves. Once large enough to leave the pouch, the joey rides on its mother's back, and stays with her until her next joey is born.

The main threats koalas face today are habitat loss and fragmentation due to logging and increasing urbanization—an estimated 4,000 koalas a year are killed by dogs and car accidents.

△ TIME TO REST

Eucalyptus leaves provide little energy, so when koalas are not sleeping up to 18 hours a day, they rest between bouts of feeding, cushioned on a dense pad of rump fur, to conserve energy.

< FAVORITE FOOD

A koala's diet is made up almost entirely of eucalyptus leaves, but it occasionally eats tea tree and wattle leaves as well.

↔ 26–32 in (65–82 cm)

⚖ 9–33 lb (4–15 kg)

⊗ Common

🌿 Eucalyptus leaves

🏠 🌳



E. Australia

Koalas eat such large quantities of **eucalyptus leaves** that their body oil **smells like cough drops**